H~M Trading Institute

QuantLab — The Quant Research OS

CONFIDENTIAL INVESTOR & TECHNICAL BRIEF — revised 2026-08-22

AI-assisted quantitative research for African retail investors — subscription SaaS, Nigerian entity, production-grade engineering.

Prepared: August 2026 · Status: Phase 0 complete, MVP production-live · Contact: mustaphahabib270@gmail.com · hmtrade.business

Confidential — prepared for investor & technical-partner review. Figures marked ≈ are indicative planning estimates; the full financial model is reproduced in Section 15 with all assumptions stated.

Executive Summary — Why This Matters Now

Africa's retail investors are the last major market with access to execution but no access to research. Brokerages let them buy; nothing helps them think. Every institutional-grade research tool is priced in USD, built for Western rails, and closed to them — while the AI that made research cheap to operate is available per-token to anyone with an API key. H~M Trading Institute closes that gap with QuantLab: a production-grade, multi-user research platform that is engineered, deployed, and running today.

What is live right now (verified 2026-08-22):

- MVP run loop verified end-to-end — prompt → queued → claim → engine → artifacts → completed, with live progress streaming (React SPA on Vercel, Python worker on Railway, Supabase backend).
- Shadow Account LIVE — the full trade-journal → behavior profile → shadow backtest → HTML report path, Premium tier. Migration applied 2026-08-22; two worker integration bugs found and fixed en route (BUG-ENG-4, BUG-ENG-5), 83 hermetic tests green.
- Billing test-ready — Paystack NGN checkout, subscription webhooks, HMAC-SHA512 verification; E2E harness 8/9 PASS, 0 FAIL (test mode).
- Admin dashboard shipped — suspension gates, user management, audit trails, database-level enforcement (10 RPCs).
- BYOK (Bring-Your-Own-Key) model — zero LLM cost-of-goods-sold; customers run unlimited research on their own DeepSeek key; platform never proxies or marks up model calls.
- Monitoring live — Sentry frontend + worker, structured JSON logs, /health endpoint.

The structural unlock: a ~$15/month infrastructure footprint, >98% gross margin, and a community-led GTM pattern that built African fintech — creator channels, trading communities, referral loops. This document is written so an investor or technical partner can understand the system end-to-end: market problem, current MVP state, architecture, roadmap, and the exact financial model (Section 15) for a $50,000 pre-seed that funds 12 months of runway.

1. The Problem — The Research Gap

Retail investors in Nigeria and across emerging markets face a research gap. Professional-grade trading research — quantitative alpha screening, multi-asset backtesting, news-and-data synthesis — is either locked inside institutions, priced in USD for Western salaries, or buried inside fragmented tools built for US/EU infrastructure (USD pricing, foreign card rails, no local context).

Meanwhile, tools that do exist for African retail (fractional brokerage, investment apps) are execution platforms. They let users buy but do not help them think. The result: self-directed traders make decisions on sentiment, with no accessible research layer — exactly when AI made such a layer cheap to operate.

Two structural facts make this solvable today: (1) frontier LLMs are available per-token to anyone with an API key, and (2) a BYOK model lets a platform operator offer unlimited AI research without bearing model cost. The customer pays for intelligence; the platform sells the product around it.

2. The Solution — QuantLab (MVP, live)

A web application where a user signs up, connects their own DeepSeek API key, and launches AI trading-research agents: natural-language prompts become runs that scan markets, backtest strategies, analyze uploaded data, and return rendered artifacts (charts, tables, reports) with live progress streaming. Three subscription tiers gate capability depth, not usage — all tiers are unlimited runs.

Tier	Price (NGN/mo)	≈USD/mo	Capabilities
Starter	₦20,000	$20,000	Single-agent research runs, unlimited
Pro	₦35,000	$35,000	Starter + Swarm orchestration (30 presets)
Premium	₦75,000	$75,000	Pro + attachments + Shadow Account (journal analysis)

Users pay DeepSeek directly for tokens (typically a small monthly LLM bill). The platform charges a flat prepaid subscription for the research product around it. A soft safety net of 30 runs/hour/user prevents abuse without creating a business quota.

2.1 Why BYOK (Bring-Your-Own-Key)

- Zero LLM margin risk — the platform never proxies or marks up model calls.
- Trust — user keys are encrypted in Supabase Vault; the platform never sees or spends them.
- Unlimited is true — since tokens are the customer's, 'unlimited runs' is a genuine statement.
- Simplified compliance — no handling of customer LLM content through our payment flow.

2.2 Shadow Account (live 2026-08-22)

A first-class Premium run kind: the user uploads a broker trade journal (CSV/XLSX/JSON), and the engine profiles their behavior, extracts an explicit 'shadow strategy', backtests it against their actual trades, and renders an HTML report. Migration applied to production 2026-08-22; the worker path was proven end-to-end via a live smoke run, during which two latent integration bugs were found and fixed (Sections 6.3, 11).

3. Why This Infrastructure — Capital-Efficient Design

No traditional backend server at MVP: Supabase (PostgreSQL 17 + Auth + Storage + Vault + Realtime + Edge Functions) is the entire backend. RLS and SECURITY DEFINER RPCs enforce access in the database — the most audit-friendly security model available.

- Paystack only, NGN — Nigerian entity; Stripe cannot onboard Nigerian businesses. Hosted checkout, recurring plans, signed webhooks (HMAC-SHA512) handle billing end-to-end. Stripe is designed for (provider-agnostic model) but deferred until the entity/Atlas path is real (R1-Q6).
- Python polling worker — runs are claimed with FOR UPDATE SKIP LOCKED: atomic, no Redis, safe with N workers. Each run executes in its own subprocess (engine uses process-global singletons).
- Cloudflare at the edge — zone, DNS, WAF, SSL (Full strict) live on hmtrade.business; R2 + Queues + Workers API gateway are the Phase 1+ target. Cloudflare Workers cannot run the engine (V8 isolates), so compute stays on Railway (D15).
- BYOK decouples model economics — see §2.1.
- Observability from day one — Sentry on frontend + worker, structured JSON logs with run IDs, /health endpoint.

Result: a production-grade, multi-tenant research platform whose monthly infrastructure cost is ~$12 before scale — the margin structure an investor can underwrite.

4. System Architecture — End-to-End View

Layer	What it is
Frontend	React 19 + Vite SPA (25+ pages: public, user, admin surfaces) — on Vercel (hmtrade.business)

Auth	Supabase Auth (email/password, JWT, branded templates)
Data	PostgreSQL 17 with RLS on every table, SECURITY DEFINER RPCs, billing/admin/audit tables
Storage	Supabase Storage (agent-artifacts, agent-uploads buckets, RLS-scoped)
BYOK Vault	Supabase Vault stores encrypted keys (service-role only decrypts)
Billing	Paystack NGN hosted checkout, recurring plans, signed webhooks
Worker	Python polling loop (Railway), FOR UPDATE SKIP LOCKED claim, subprocess-per-run isolation
Engine	Tradi (vendored Vibe-Trading 0.1.14+): ReAct agent, 462 alphas, 8 backtest engines, 24 data sources, 78 tools, 90 skills, 30 swarm presets, shadow-account toolset
LLM	DeepSeek via BYOK — never proxied or marked up
Observability	Sentry (frontend + worker), structured JSON logs, /health
Edge	Cloudflare: zone hmtrade.business, proxied DNS, WAF, Full-strict SSL

4.1 The Run Pipeline (core flow)

1. User launches a run; the start_agent_run RPC validates auth → active subscription → DeepSeek key configured.
2. Row inserted as 'queued' with a unique idempotency key; Realtime notifies the browser.
3. Worker polls WHERE status='queued' with FOR UPDATE SKIP LOCKED, claims one row.
4. Worker spawns a subprocess: vibe-trading run -p '<prompt>' --json, with per-run isolated HOME.
5. Progress streams back; artifacts upload to Storage.
6. Run completes/fails; the UI renders artifacts via Realtime.

A run is a row, a claim, a subprocess and a stream — nothing more. That simplicity is the design's resilience.

4.2 Provenance & the research substrate (Phase 1, planned)

The Phase 1 brief (docs/PHASE1_BRIEF.md) stands: a FastAPI modular monolith on the same Railway container (two processes), adding projects/hypotheses/experiments/dataset_registry/promotion_events tables + RLS, worker-side provenance capture on every run, Supabase JWT validation, and an Upstash Redis skeleton (health + queue smoke, fail-open). The live run loop never breaks (D16 strangler-fig).

5. Data Strategy — Canonical Layers (DATA.md)

Market data is treated as a versioned product and scientific dependency, not an API response (DATA.md). The canonical pipeline:

Layer	Meaning
RAW	As-received, never overwritten
VALIDATED RAW	Sequence/gap validation
NORMALIZED	Canonical schema, multiple providers → one model
DERIVED	Computed fields, corporate actions
FEATURES	Versioned feature layer (training-serving parity)

Every dataset carries metadata: provider, venue, symbols, asset class, frequency, timezone, timestamp semantics, coverage, adjustment/corporate-action policy, point-in-time capability, license, version, quality score. Point-in-time is a hard invariant: historical experiments may only use information knowable at the simulated time (event time + knowledge time). Phase 2 targets a Supabase-Storage lake (Parquet) with DuckDB/Polars access and a dataset registry; the R2 billing decision is deferred until real data volume exists (R2-Q8). First asset class: crypto (R1-Q7), first dataset BTC/ETH/SOL daily OHLCV (R3-Q4), reusing the engine's OKX/Binance/CCXT loader fallbacks.

6. Research Workflow (WORKFLOW.md)

The research pipeline the platform is being built toward (the visual form lives in WORKFLOW.md and INFRASTRUCTURE.md):

7. User question → AI research agent → formal hypothesis → experiment spec → data requirements → data snapshot → feature pipeline
8. Baseline / ML / RL experiments → backtest engine → cost & slippage → walk-forward validation → untouched test set
9. Pass/fail → promotion ladder: candidate → validated → paper → shadow → approved → live (Phase 4 workflow; the promotion vocabulary is already in the Phase 1 schema, per RECONCILIATION R12)
10. Research feedback loop: results feed the next question — an explicit, governed loop, not a black box.

Engineering workflow: every change rides the same discipline — 83 hermetic worker tests, frontend build + 261/265 vitest, engine suite, CI grep gates, and the launch checklist as the single tracking artifact.

7. UI Vision — QuantLab & QuantLab Admin (UI_VISION.md)

Brand lock (2026-08-21): customer-facing brand H~M Trading Institute; product QuantLab (user app) + QuantLab Admin (control plane). One design system, one palette, one typography — locked, never alternatives (design-system rule).

QuantLab (user app) — data-dense, calm, professional, AI-native: Overview/Dashboard (portfolio, P&L, Sharpe, drawdown — never fake numbers, R17), Markets, Research (AI copilot prompt + research cards), Strategies, ML Studio, RL Studio, Backtests, Paper trading, Portfolio, Data, Models, Live. Live controls are visually distinct from research/paper surfaces.

QuantLab Admin (control plane) — platform operations: user management, subscription/plan overrides, suspension controls, audit logs, system health. The current /admin SPA surface stays; RBAC granularity + MFA (TOTP) are roadmap items (R10).

8. Current Status — What Is Live Today (2026-08-22)

Component	Status
Frontend SPA	✓ LIVE: React 19 + Vite on Vercel (hmtrade.business / hm-ashy-six.vercel.app)
Auth	✓ LIVE: Supabase Auth + branded templates (dashboard paste pending)
Run pipeline	✓ VERIFIED: prompt → queued → claim → engine → artifacts → completed; BUG-ENG-4/5 fixed 2026-08-22
Billing (Paystack)	✓ TEST MODE: E2E harness 8/9 PASS; test-card flow verified; live keys pending at launch
BYOK + Vault	✓ LIVE: encrypted key storage, worker decrypts server-side
Swarm (30 presets)	✓ LIVE: Pro/Premium gate, user_vars routing
Attachments (CSV/XLSX/JSON)	✓ LIVE: Premium tier, 50 MB cap, Storage upload/RLS
Shadow Account	✓ LIVE: journal → profile → shadow backtest → HTML report (Premium); migration + fixes 2026-08-22
Admin dashboard	✓ LIVE: suspend/unsuspend, user list, plan override, audit logs (10 RPCs, DB-enforced)
Worker tests	✓ 83 hermetic tests, black-clean, all passing
Monitoring	✓ LIVE: Sentry frontend + worker, structured JSON logs, /health
Cloudflare + SSL	✓ LIVE: zone hmtrade.business, proxied DNS, Full strict, GTS cert
Railway worker	✓ LIVE: service HM, /health :9100, WORKER_EXECUTE_TRADI=true
Email	✓ Templates shipped (branded); support@ mailbox + Resend SMTP pending
Keepalive + backup	✓ SHIPPED: daily keepalive + weekly pg_dump workflows; Actions +

	DATABASE_URL secret pending

The run loop is production-verified. Billing is test-ready. Admin is live. Remaining before launch: Paystack live key switch + charge-case E2E, Auth URLs + branded emails (Resend), support mailbox, legal fields, test-account cleanup, env-var purge.

9. Roadmap — The Quant Research OS (10 Phases)

The destination: from a single-purpose AI research wrapper into a full Quant Research OS. Each phase stands up beside the live system (strangler fig), never breaking the run loop (D16).

Phase	Name	Status
0	Baseline & freeze (inventory, branch strategy, constitution)	✓ DONE 2026-08-21
1	Foundation — FastAPI monolith, tenant projects, research-governance substrate (projects, hypotheses, experiments, dataset_registry, promotion_events; provenance capture; Redis skeleton fail-open)	⏳ POST-LAUNCH (branch upgrade/phase-1-foundation)
2	Data — canonical schema, PIT, Parquet lake (Supabase Storage), dataset registry, BTC/ETH/SOL first	planned
3	Quant Engine — strategy SDK, ExecutionInterface, walk-forward, realistic backtesting	planned
4	Research AI — pre-registration workflow, hypotheses, experiment workflow, promotion ladder	planned
5	ML — baseline first, model registry (MLflow candidate)	planned
6	Paper Trading — risk engine, OMS, reconciliation, WebSocket-first plane (D17)	planned
7	RL — research-only, isolated environments	planned
8	Controlled Live — broker adapters, mandate-gated, fail-closed OMS	planned
9	Scale — observability,	planned

	metered billing, quotas, cost attribution ($150–250/mo target)	
10	Expansion — asset classes, public API, mobile views, marketplace, enterprise	planned

Sequencing: Phase 0 ✓ DONE. Nigeria MVP launch now. Phase 1 begins post-launch (R1-Q1). Governance decisions locked in UPGRADE_ROADMAP.md §4 (R1–R3 grilling): launch-first, Hermes-executed Phase 0, feature branches per phase, same-container two-process FastAPI, Upstash Redis free-tier skeleton, Stripe deferred, crypto first, minimal projects table, Supabase free + keepalive + pg_dump, provenance captured by both worker and FastAPI, capture-first (pre-registration UX = Phase 4), Railway $5 through launch (evaluate Fly.io free at Phase 1), research-domain-only first FastAPI surface, Resend free tier for email.

10. Reconciliation — Unified Spec Authority

docs/UNIFIED_SPEC.md (63 sections, adopted 2026-08-21) is the single source of truth for the target architecture; docs/RECONCILIATION.md records the 18 conflicts with existing ADRs/live stack and their resolutions; docs/REQUIREMENTS.md is the 81-ID requirements register (13 groups, MUST/SHOULD/MAY, traceable to roadmap phases). Key reconciliations:

#	Unified spec says	Resolution
R1	Next.js frontend	Keep D3 React 19 + Vite for the live product; Next.js is a future refactor decision
R2	AWS primary cloud	Keep Railway + Supabase + Cloudflare until scale justifies AWS (Phase 9/10)
R3	Stripe billing	Keep D8 Paystack NGN first; Stripe activates when entity exists; spec's provider abstraction matches D8
R4	Celery/RQ via Redis	Keep D14 SQL SKIP LOCKED for MVP; queue enters Phase 1 with FastAPI
R5	LLM gateway/router	BYOK+Vault stays the key mechanism (D12); gateway is Phase 3+
R6	MLflow	Adopt as Phase 5 candidate implementation; our provenance schema stays canonical
R7	Redis cache/rate-limit	Keep Postgres rate limit (30/hr) until Phase 1 Redis skeleton
R10	Admin RBAC/MFA	Keep SPA /admin; add roles

		+ TOTP as roadmap items
R11	WebSocket-first	D17 recorded; REST + polling stay until Phase 6 WS plane
R12	Registry lifecycles	Spec lifecycles refine the Phase 1 DDL (DRAFT→… →RETIRED; candidate→… →live)
R13	Repo layout apps/services	Target layout for new code only; no big-bang restructure (D16)
R17	No fake numbers	Adopted — empty states + paper-account offer

11. Key Architecture Decisions (ADRs)

ID	Decision	Rationale
D1	Subprocess-per-run	Engine uses process-global singletons; in-process multi-tenant is unsafe
D3	React 19 + Vite frontend	Saves a rewrite; inherits Vibe-Trading alpha UI
D5	SQL job claiming (FOR UPDATE SKIP LOCKED)	Atomic, no Redis, N workers safe
D8	Paystack NGN first (Stripe planned)	Nigerian entity; provider-agnostic subscription model
D11	BYOK pivot	Users supply keys; unlimited runs per tier; zero LLM margin risk
D12	Supabase Vault for key storage	Encryption without app-managed secrets
D13	Edge Functions for billing	No FastAPI backend at MVP
D14	Python polling loop, no Celery/Redis	Simpler; sufficient for MVP scale; queue enters Phase 1
D15	Railway + Supabase + Cloudflare	Cost-efficient compute/data/edge; CF Workers cannot run the engine
D16	Strangler-fig migration	Live loop never breaks; new code beside old
D17	WebSocket-first commitment	Market data, orders, fills, P&L, job progress over WS (Phase 6)

11.1 Verification gates

- Worker: 83 hermetic tests (pytest, black-clean).
- Frontend: npm run build + 261/265 vitest (4 pre-existing Layout flake).

- Engine: pytest suite, upstream 0.1.14+ (1907e47) with DeepSeek reasoning absorption.
- Billing: E2E harness (offline fixtures + live mode, hard sk_live_ guard), 8/9 PASS.
- Webhook: Paystack HMAC-SHA512 constant-time verification + API re-verification.
- Auth: signUp → signIn → protected route → signOut happy path.
- BYOK: save key → Vault → worker decrypts → run completes.

12. Repository & Testing

Path	Role
Tradi/frontend/	React SPA — production frontend (public + user + admin routes)
Tradi/	Vendored Vibe-Trading engine (MIT) — agent, backtests, data sources, skills
vibe-trading-saas/worker/	Python worker — polling loop, runner, artifacts, logging, health
vibe-trading-saas/db/migrations/	SQL migrations (applied manually to Supabase)
supabase/functions/	Edge Functions: paystack-init, paystack-webhook
supabase/email-templates/	Branded auth + transactional email templates
scripts/	E2E harness + template validator (zero-dependency Node)
CLAUDE.md	Single source of truth for the live system: schema, RPCs, sprint tracker
SOUL/PROJECT/DATA/ARCHITECTURE/ WORKFLOW/FOUNDATIONS.md	Constitution — mission, identity, data, target architecture, workflows
UPGRADE_ROADMAP.md	Harmonized 10-phase migration plan + decision ledger
docs/	UNIFIED_SPEC.md, RECONCILIATION.md, REQUIREMENTS.md, PHASE1_BRIEF.md, LAUNCH_CHECKLIST.md, UI_VISION.md, INFRASTRUCTURE.md

13. Security & Compliance Posture

- Row isolation: RLS on every user-facing table (user_id = auth.uid()).
- BYOK keys: Supabase Vault; public schema holds only opaque secret_id; decryption service-role only.
- Auth gate: Supabase Auth JWT on every protected route (AuthGuard + AdminGuard).
- Subscription gate: start_*_run RPCs check active sub + configured key server-side.
- Webhooks: Paystack x-paystack-signature HMAC-SHA512 constant-time + API re-verify; idempotency via webhook_events UNIQUE.
- Run isolation: subprocess gets own HOME / VIBE_TRADING_HOME / VIBE_TRADING_ALLOWED_RUN_ROOTS; worker secrets stripped from engine env.
- Artifact access: Supabase Storage, owner-scoped RLS, signed URLs (5-min TTL).

- No live trading: mandate-gated, off by default at MVP (SOUL principle #6).
- Fail closed: uncertain broker state / stale data / recon mismatch / risk breach blocks unsafe actions (SOUL #8).

14. Business Model & Unit Economics

Revenue: flat monthly NGN subscription (prepaid via Paystack). No usage metering, no token markup. COGS: customer's own DeepSeek spend (outside the platform). Infra: serverless Supabase + one small Railway worker ≈ $12/mo. Structurally >98% gross margin at scale.

Metric	Value
Blended ARPU	₦30,000 / $20
Paystack effective fee	1.83% (1.5% + ₦100/txn at blended ARPU)
Gross margin	98.2%
Lifetime value (LTV)	$286 at 7%/mo churn
Target CAC	$8 (community-led GTM)
LTV:CAC	35.7:1
CAC payback	0.4 months
Monthly infra (MVP)	$12

Acquisition: creator/community-led GTM (trading communities, educator channels) — the pattern that built African fintech. Expansion: public API + mobile views open B2B2C (brokers, educators reselling research); Stripe enables pan-African once the entity permits.

15. Financial Pitch — Budget, Model & Funding Ask

15.1 The ask

We are raising $50,000 in pre-seed funding — a 12-month runway at a $4,200/month burn — to launch the product we have already built and take it from zero to ~750 paying subscribers ($180,000 ARR) within a year.

15.2 Revenue model & scenarios

Scenario	Subscribers	MRR (≈USD)	ARR (≈USD)	Net after fees & infra (≈/mo)
S100	100	$2,000	$24,000	$1,951
S250	250	$5,000	$60,000	$4,896
S500	500	$10,000	$120,000	$9,805
S1000	1000	$20,000	$240,000	$19,621

Planning exchange rate: ₦1,500/USD. Blended ARPU (60% Starter / 30% Pro / 10% Premium) = ₦30,000 ≈ $20/mo. Paystack local-card fee 1.5% + ₦100 per charge. Gross margin is structurally >98% because LLM tokens are the customer's own (BYOK).

15.3 12-month ramp (conservative, community-led)

Month	Paying subs (EOM)	MRR (≈USD)	ARR (≈USD)
M1	25	$500	$6,000

M2	60	$1,200	$14,400
M3	110	$2,200	$26,400
M4	170	$3,400	$40,800
M5	240	$4,800	$57,600
M6	320	$6,400	$76,800
M7	400	$8,000	$96,000
M8	470	$9,400	$112,800
M9	540	$10,800	$129,600
M10	610	$12,200	$146,400
M11	680	$13,600	$163,200
M12	750	$15,000	$180,000

Break-even: ~210 subscribers (≈$4,200 MRR) covers the $4,200/month burn — reached around month 5 on the ramp above.

15.4 Use of funds — $50,000 budget

Category	Share	Amount	What it buys
Go-to-market & acquisition	40%	$20,000	Creator partnerships, trading-community campaigns, referral incentives, content engine
Compliance, legal & entity	20%	$10,000	Registrant block, ToS/Privacy enforceability, data-protection (NDPR) posture, support infrastructure
Product expansion	25%	$12,500	Phase 1 FastAPI + research governance, public API, mobile views, international billing (Stripe), admin 2FA
Working capital & runway	15%	$7,500	Paystack settlement float, infra buffer at scale, contingency

15.5 Unit economics summary

Metric	Value	Why it matters
Gross margin	>98%	BYOK means no LLM COGS; the platform's margin is software margin
LTV:CAC	35.7:1	Community-led GTM keeps CAC near-zero; healthy SaaS benchmark is ≥3:1

Payback	0.4 months	Cash returned on acquisition in under half a month
Infra cost per 1,000 users	≈$12–40/mo	Serverless stack scales cheaply; compute only grows with Phase 2 data
Runway from raise	11.9 months	Room to reach break-even (~month {F['be_month']}) and iterate

15.6 What the money is NOT for

The engineering is done — the product is built, deployed, and verified. The raise funds distribution, compliance, and expansion — not further engineering of the core loop. That is the underwriting thesis: capital unlocks the company (distribution + compliance), not the product.

"The engineering is done. The company is what the raise builds."

16. Why We Need Investors — Three Concrete Gaps

- Distribution: a research product with zero marketing budget is undiscovered. Retail trading in Nigeria is won on the ground (creator channels, trading communities, referral loops) — that costs money before it compounds.
- Compliance & entity cost: operating a paid financial-research service in Nigeria requires legal review, registrant identity (physical address, business name), data-protection posture (NDPR), support infrastructure. Fixed, non-optional costs.
- Working capital & expansion: Paystack settlement float, compute scaling, public API, mobile views, international (Stripe) billing all need cash runway — not more engineering.

Appendix A — Phased Delivery Record

Phase & deliverable	Status (2026-08-22)
Phase 0–1: Auth, dashboard, agent launcher, run view	✓ SHIPPED
Phase 2: Worker E2E (claim → subprocess → artifacts)	✓ SHIPPED & VERIFIED
Phase 3: BYOK pivot (Vault, RPCs, worker decrypt)	✓ SHIPPED
Phase 4: Swarm (30 presets, Pro/Premium gate) + attachments	✓ SHIPPED
Phase 5: Realtime queue, progress streaming, billing Edge Functions, pricing, legal	✓ SHIPPED
Phase 6: Admin dashboard (10 RPCs, suspension gate, audit trail)	✓ SHIPPED & LIVE
Phase 7: Monitoring (Sentry, structured	✓ LIVE

logs, /health)	
Paystack E2E (hermetic harness, webhook hardening)	✓ 8/9 GREEN (test mode)
Shadow Account (journal → profile → backtest → report)	✓ SHIPPED & LIVE (2026-08-22)
Engine update 0.1.14+ + worker fixes (BUG-ENG-4/5)	✓ LIVE (2026-08-22)
Cloudflare zone + SSL	✓ LIVE
Railway worker deploy (production env)	✓ LIVE
Keepalive + backup workflows	✓ SHIPPED (Actions + secret pending)
Launch prep (Paystack live switch, emails, mailbox, legal, cleanup)	⏳ PENDING

Appendix B — Cost Structure & Timeline

B.1 Monthly infrastructure cost

Component	Plan	Cost
Vercel (frontend)	Free	$0
Supabase (Postgres, Auth, Storage, Vault, Realtime)	Free + keepalive	~$0 (free tier)
Cloudflare (DNS, WAF, edge)	Free	$0
Railway (worker container)	Starter	$5
Paystack (billing)	Test → Live	$0 test + 1.5% + ₦100/txn live
Resend (email, Phase 1)	Free (3,000/mo)	$0 → $20 at scale
Sentry (monitoring)	Developer	$0 (free tier)
Estimated total MVP		~$12–15/mo before scale

B.2 Timeline

Window	Milestone
Now (2026-08-22)	Phase 0 complete; MVP production-live; Shadow Account live; run loop verified after BUG-ENG-4/5
Weeks 1–2	Launch: Paystack live switch, charge-case E2E, support@ mailbox, legal review, acquisition cohort
Weeks 3–8 (R1-Q1)	Monitor live: churn, CAC, support load; Phase 1 branch begins (FastAPI + research governance)
Weeks 9–16	Phase 1 deployment (code + manual SQL); Phase 2 scoping (data lake, PIT, canonical schema)
Q4 2026 onward	Phases 3–10: quant engine, research AI, ML/RL, paper, live, scale, expansion

Appendix C — Cold Email (short, attention-grabbing)

Subject: Africa's retail investors have execution but no research. We built the research layer.

Hi [Name],

Africa's retail investors can buy stocks — but nothing helps them think. Every serious research tool is priced in USD and built for Western rails. We built the missing layer.

QuantLab is a production-grade, AI-native research platform for African retail traders — live today, with a verified run loop, Paystack billing, and a BYOK model that gives us >98% gross margin. Users bring their own API key; we sell the research product around it. No LLM COGS, ~$12/month infra, 12-month runway on a $50k pre-seed.

Would you have 20 minutes this week to see the product and the model? Happy to share the full brief.

Best,
Musty — H~M Trading Institute · hmtrade.business